

Project Initialization and Planning Phase

Date	29 June 2026
Team ID	
Project Title	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to develop OptiCrop, a machine learning-based crop recommendation system that improves crop selection using soil and environmental parameters. It addresses inefficient farming decisions by providing accurate crop recommendations, leading to better productivity and resource utilization. Key features include a machine learning prediction model, an interactive web application, and real-time crop recommendations.

Project Overview	
Objective	Develop an AI-powered crop recommendation system that analyzes soil nutrients (Nitrogen, Phosphorus, Potassium, pH) and environmental conditions (temperature, humidity, rainfall) to recommend the most suitable crop, improving agricultural productivity and resource utilization.
Scope	The project provides farmers with accurate crop recommendations using machine learning. It includes soil and weather data analysis, a web-based interface for predictions, and supports future integration with IoT sensors and weather APIs.
Problem Statement	
Description	Farmers often rely on traditional farming practices and personal experience when selecting crops. Variations in soil nutrients and changing weather conditions frequently lead to poor crop selection, lower yields, unnecessary fertilizer usage, and financial losses.
Impact	An incorrect crop choice reduces agricultural productivity, increases cultivation costs, wastes natural resources, and negatively affects farmers' income and long-term sustainability.
Proposed Solution	
Approach	Develop a machine learning-based crop recommendation system using a Random Forest classifier. The system processes soil and weather parameters to predict the most suitable crop through an easy-to-use Flask web application.
Key Features	1) AI-based crop recommendation using Random Forest. 2) Soil parameter analysis (N, P, K, pH). 3) Weather parameter analysis (Temperature, Humidity, Rainfall). 4) High prediction accuracy (99.55%). 5) User-friendly Flask web interface. 6) Fast real-time crop prediction.

	7) Future support for IoT sensors and Weather API integration.
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ResourceRequirements

ResourceType	Description	Specification/Allocation
Hardware		
Computing Resources	CPU specifications	Intel Core i5 (or equivalent)
Memory	RAM	8 GB or higher
Storage	Disk space	20 GB free SSD/HDD
Software		
Frameworks	Backend Framework	Flask
Libraries	ML & Data Processing	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Development Environment	IDE	Jupyter Notebook, VS Code or JupyterLab
Data		
Data	Source, size, format	UCI Crop Recommendation Dataset, CSV format (~2200 records)